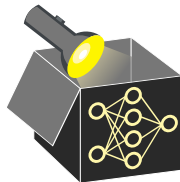
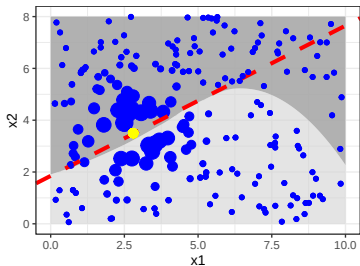


Interpretable Machine Learning



Local Explanations: Lime

Local Interpretable Model-agnostic Explanations (LIME)



Learning goals

- Understand motivation for LIME
- Develop a mathematical intuition

LIME

- **Locality assumption:**

$\hat{f}$ behaves similarly simple in a small neighborhood of $\mathbf{x}$

~> Approximate $\hat{f}$ near $\mathbf{x}$ using an interpretable surrogate model $\hat{g}$

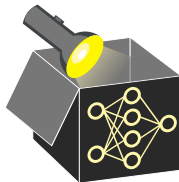
- **Interpretation strategy:**

Use $\hat{g}$'s simple internal structure to explain $\hat{f}(\mathbf{x})$ locally

~> **Common surrogates:** sparse linear models, shallow decision trees

- **Applicability:** Model-agnostic; supports tabular, image, and text data

- **In practice:** Generate samples near $\mathbf{x}$, predict with $\hat{f}$, and fit $\hat{g}$ to these samples using $\hat{f}$'s outputs as targets, weighting samples by their proximity to $\mathbf{x}$



LIME: CHARACTERISTICS

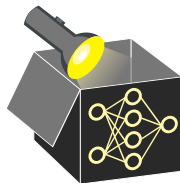
Definition:

LIME provides a local explanation for a black-box model $\hat{f}$ in form of a surrogate model $\hat{g} \in \mathcal{G}$, where $\mathcal{G}$ is a class of interpretable models

Surrogate model $\hat{g}$ should satisfy two characteristics:

- 1 **Interpretable:** Provide human-understandable insights into the relationship between input features and prediction, e.g. via coefficients or model structure
- 2 **Local fidelity / faithfulness:** $\hat{g}$ closely approximates $\hat{f}$ in the vicinity of the input $\mathbf{x}$ being explained

Goal: Find $\hat{g}$ with **minimal complexity and maximal local fidelity**



MODEL COMPLEXITY

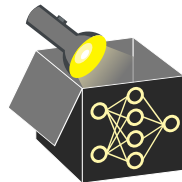
We can measure complexity of $\hat{g} \in \mathcal{G}$ using a complexity measure $J : \mathcal{G} \rightarrow \mathbb{R}_0$

- **Example: (Sparse) Linear Models**

Let $\mathcal{G} = \{g : \mathcal{X} \rightarrow \mathbb{R} \mid g(\mathbf{x}) = s(\boldsymbol{\theta}^\top \mathbf{x})\}$ be the class of linear models

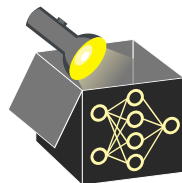
- $s(\cdot)$ is the identity (linear model) or logistic sigmoid function (logistic regression)

$\rightsquigarrow J(g) = \sum_{j=1}^p \mathbb{1}_{\{\theta_j \neq 0\}}$: count the number of non-zero coefficients via the L_0 norm of $\boldsymbol{\theta}$



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- **Example: Decision Trees**

Let $\mathcal{G} = \{g : \mathcal{X} \rightarrow \mathbb{R} \mid g(\mathbf{x}) = \sum_{m=1}^M c_m \mathbb{1}_{\{\mathbf{x} \in \mathcal{R}_m\}}\}$ be the class of trees

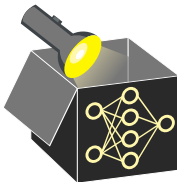
- $\mathcal{R}_m$ are disjoint axis-parallel regions (leaves); $c_m \in \mathbb{R}$ are constant predictions

$\rightsquigarrow J(g) = M$: count the number of terminal or leaf nodes

LOCAL FIDELITY OF SURROGATE MODELS

- Surrogate $\hat{g}$ is **locally faithful** to a black-box model $\hat{f}$ around an input $\mathbf{x}$ if

$$\hat{g}(\mathbf{z}) \approx \hat{f}(\mathbf{z}) \quad \text{for synthetic samples } \mathbf{z} \in \mathcal{Z} \subseteq \mathbb{R}^p \text{ generated around } \mathbf{x}$$

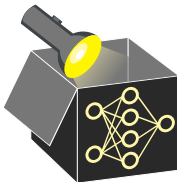


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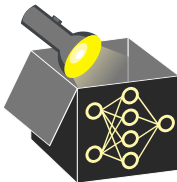
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 - ❶ **A proximity (similarity) measure** $\phi_{\mathbf{x}}(\mathbf{z})$ between $\mathbf{z}$ and $\mathbf{x}$, e.g.:

$$\phi_{\mathbf{x}}(\mathbf{z}) = \exp(-d(\mathbf{x}, \mathbf{z})^2 / \sigma^2)$$

(exponential kernel), where

- d is a distance metric, e.g. Euclidean or Gower for mixed types
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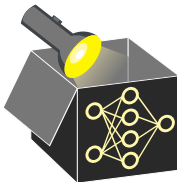
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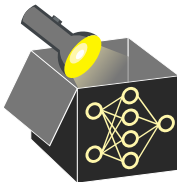
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$$L(\hat{f}(\mathbf{z}), \hat{g}(\mathbf{z})) = (\hat{g}(\mathbf{z}) - \hat{f}(\mathbf{z}))^2$$



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- The overall **local fidelity objective** is measured by a weighted loss:

$$L(\hat{f}, \hat{g}, \phi_{\mathbf{x}}) = \sum_{\mathbf{z} \in \mathcal{Z}} \phi_{\mathbf{x}}(\mathbf{z}) \cdot L(\hat{f}(\mathbf{z}), \hat{g}(\mathbf{z}))$$

LIME OPTIMIZATION TASK

- Optimization problem of LIME:

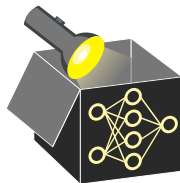
$$\arg \min_{\hat{g} \in \mathcal{G}} L(\hat{f}, \hat{g}, \phi_{\mathbf{x}}) + J(\hat{g})$$

- **In practice** LIME uses a two-stage approach:

- ➊ User sets complexity $J(\hat{g})$ beforehand, e.g. LASSO with k features
- ➋ Optimize $L(\hat{f}, \hat{g}, \phi_{\mathbf{x}})$ for fixed complexity

- **Goal:** Build a **model-agnostic** explainer

- ↪ Optimize $L(\hat{f}, \hat{g}, \phi_{\mathbf{x}})$ without making assumptions on the form of $\hat{f}$
- ↪ Surrogate $\hat{g}$ approximates f locally through sampling and fitting

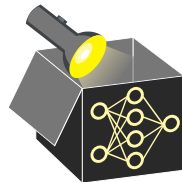


LIME ALGORITHM: OUTLINE

► "Ribeiro." 2016

Input:

- Pre-trained black-box model $\hat{f}$
- Observation $\mathbf{x}$ whose prediction $\hat{f}(\mathbf{x})$ we want to explain
- Interpretable model class $\mathcal{G}$ for the local surrogate



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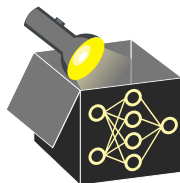
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Algorithm:

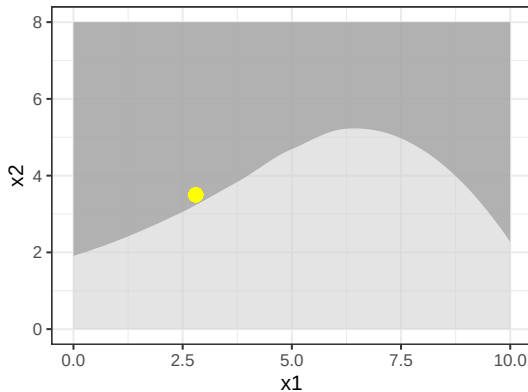
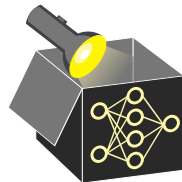
- 1 Independently sample new points $\mathbf{z} \in \mathcal{Z}$
- 2 Retrieve predictions $\hat{f}(\mathbf{z})$ for the obtained points $\mathbf{z}$
- 3 Weight $\mathbf{z} \in \mathcal{Z}$ by their proximity $\phi_{\mathbf{x}}(\mathbf{z})$ to quantify closeness to $\mathbf{x}$
- 4 Train interpretable surrogate model $\hat{g}$ on data $\mathbf{z} \in \mathcal{Z}$ using weights $\phi_{\mathbf{x}}(\mathbf{z})$
 $\rightsquigarrow$ Predictions $\hat{f}(\mathbf{z})$ are used as the target of this model
- 5 Return $\hat{g}$ as the local explanation for $\hat{f}(\mathbf{x})$



LIME ALGORITHM: EXAMPLE

Illustration of LIME based on a classification task:

- Light/dark gray background: prediction surface of a classifier
- Yellow point: $\mathbf{x}$ to be explained
- $\mathcal{G}$: class of logistic regression models



LIME ALGO.: EXAMPLE (STEP 1+2: SAMPLING)

Strategies for sampling:

- Uniformly sample new points from the feasible feature range
- Use the training data set with or without perturbations
- Draw samples from the estimated univariate distribution of each feature
- Create an equidistant grid over the supported feature range

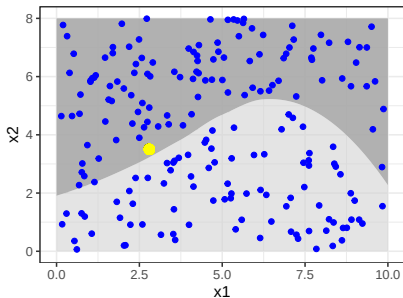
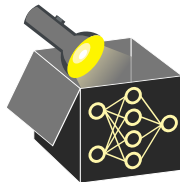


Figure: Uniformly sampled

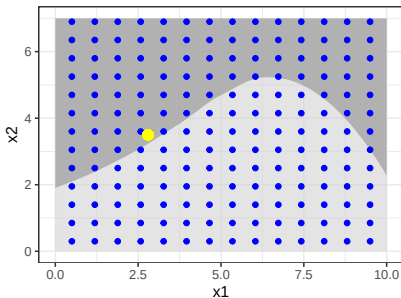
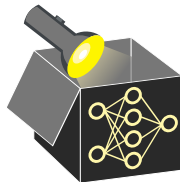
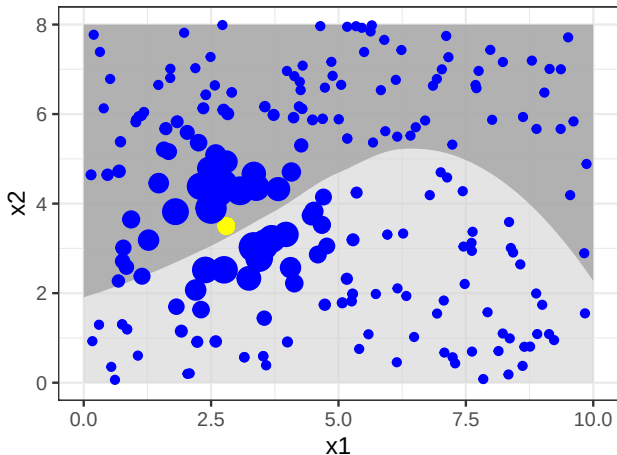


Figure: Equidistant grid

LIME ALGO.: EXAMPLE (STEP 3: PROXIMITY)

In this example, we use the exponential kernel defined on the Euclidean distance d :

$$\phi_{\mathbf{x}}(\mathbf{z}) = \exp(-d(\mathbf{x}, \mathbf{z})^2 / \sigma^2)$$



LIME ALGO.: EXAMPLE (STEP 4: SURROGATE)

In this example, we fit a **logistic regression** model

$\leadsto L(\hat{f}(\mathbf{z}), \hat{g}(\mathbf{z}))$ is the Bernoulli loss

